EXPERIMENT – 3

AIM :

To write a Python program to solve the Water Jug Problem by measuring a required quantity of water using two jugs of different capacities.

ALGORITHM:

1. Start.
2. Initialize two jugs with their capacities.
3. Set the initial water level of both jugs to zero.
4. Define the possible operations: fill, empty, and pour water between jugs.
5. Apply the operations to reach the required quantity.
6. Repeat until the goal state is reached.
7. Display the final water levels.
8. Stop.

PROGRAM :

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```
from collections import deque

def water_jug():
    q = deque([(0, 0)])
    visited = set()

    while q:
        a, b = q.popleft()

        if (a, b) in visited:
            continue

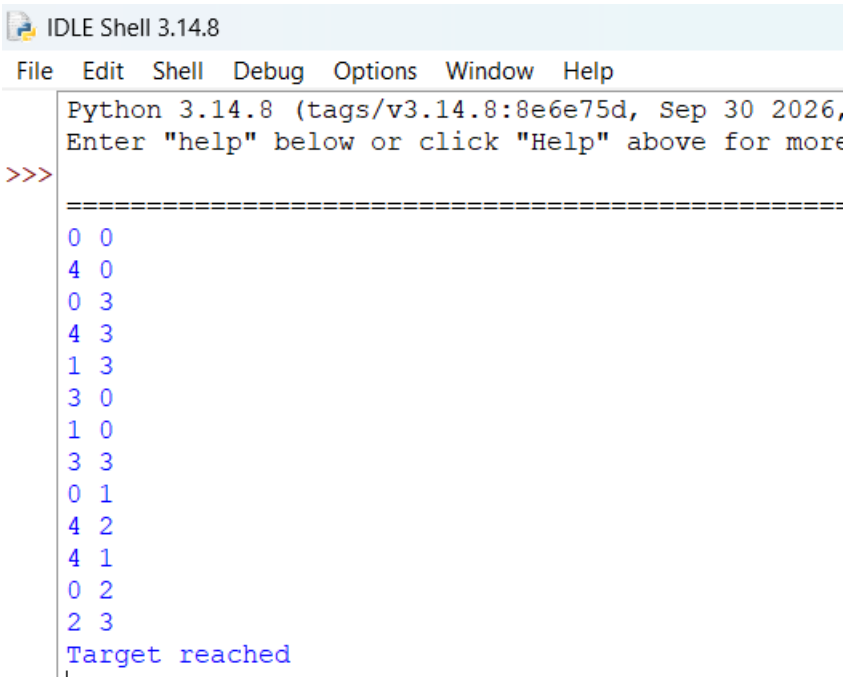
        visited.add((a, b))
        print(a, b)

        if a == 2:
            print("Target reached")
            return

        q.extend([
            (4, b), (a, 3), (0, b), (a, 0),
            (a - min(a, 3-b), b + min(a, 3-b)),
            (a + min(b, 4-a), b - min(b, 4-a))
        ])

water_jug()
```

OUTPUT:



RESULT:

Thus, the Python program to solve the Water Jug Problem was executed successfully, and the required quantity of water was obtained.